Econ 30065: Microeconomics
Sara Negrelli

Labor Supply

Labor Supply

- Consumers buy goods and services
- Many are also sellers (e.g., sell their work effort)
- ***Labor supply*** refers to the sale of a consumer's time and effort to an employer
- How to study labor supply:
  - For every bad, there is a corresponding good, defined by the absence of the bad.
  - If people regard hours of work as a bad, the corresponding good is *leisure*.
  - The price of hours of leisure is **wage**
  - **To understand supply of labor => we study the demand for leisure!**

Choice

- 2 goods: consumption good (C) e leisure time (N)
  - (labor L=T-N) where T is time
  - Standard preferences
- Budget constraint:
  - Income is given by the sum of initial endowment (M_0) and labor income
  - P_C price of C
  - Salary w is the price of leisure time
  - Budget constraint: $p_c C$=endowment+w(T-N)

Labor-Leisure Choice: An Example

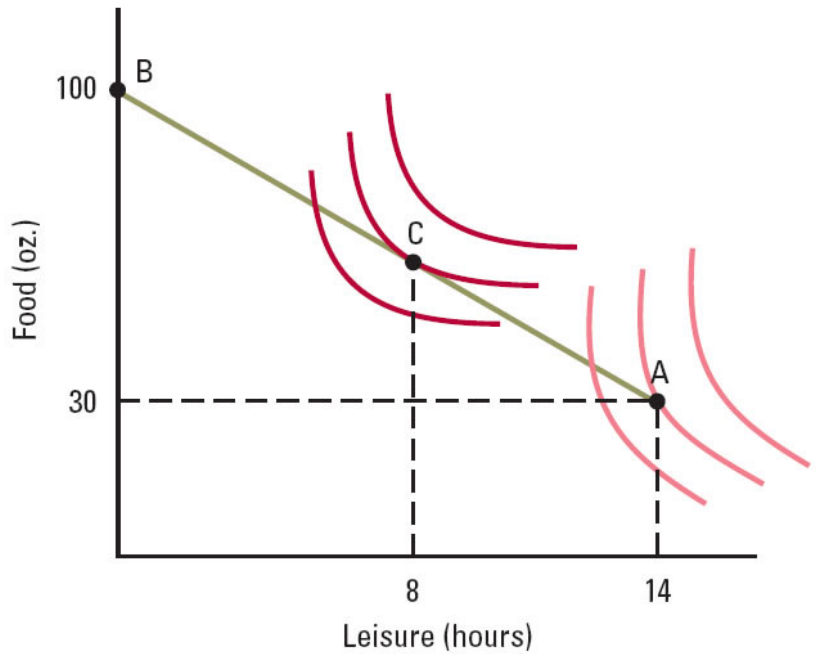
- Javier's possible income sources:
  - Endowment of \$30 per day
  - Job that pays \$5 per hour
- 14 hours per day available to allocate toward work and leisure
- Assume all money spent on C (1\$ a ounce)
- Budget constraint:

$$C=w(14-N)+\text{ endowment}$$

$$C+5N=5\text{x}14+30=100$$

Labor-Leisure Choice: An Example

- Decision about how many hours of leisure to enjoy (and thus how many to work) depends on his preferences:
  - With the dark red preferences, Javier chooses 8 hours of leisure (6 hours of work) per day
  - With the light red preferences, Javier chooses not to work



Example

Daniel is a university student. In order to earn some money, he gives math private lessons. His hourly wage w is 5. Daniel has 16 hours (T) each day to spend in labor (L) and leisure (N). The price of his consumption good is 1. His utility function is: $U(N,C)=NC$

Solution

- Write down his budget constraint:

  C +5N= 80

- Find his best choice of C and N:

  MRS= price ratio: C/N = 5

  Use the 2 equations:

  C=40, N=8

Effect of a change in wage on hours of work

- How does a change in wage affect a consumer's budget line?
  - Look at budget line pC=w(T-N), where T is the total time to be allocated between leisure and work.
  - Y-intercept increases as *w* increases
  - X-intercept does not depend on *w*
  - Rotation of the budget line, getting steeper with higher hourly wages
- Points of tangency between indifference curves and rotating budget lines form a price-consumption path
- This leads to:
  - leisure demand curve
  - labor supply curve (mirror image of leisure demand curve)

Labor supply curve

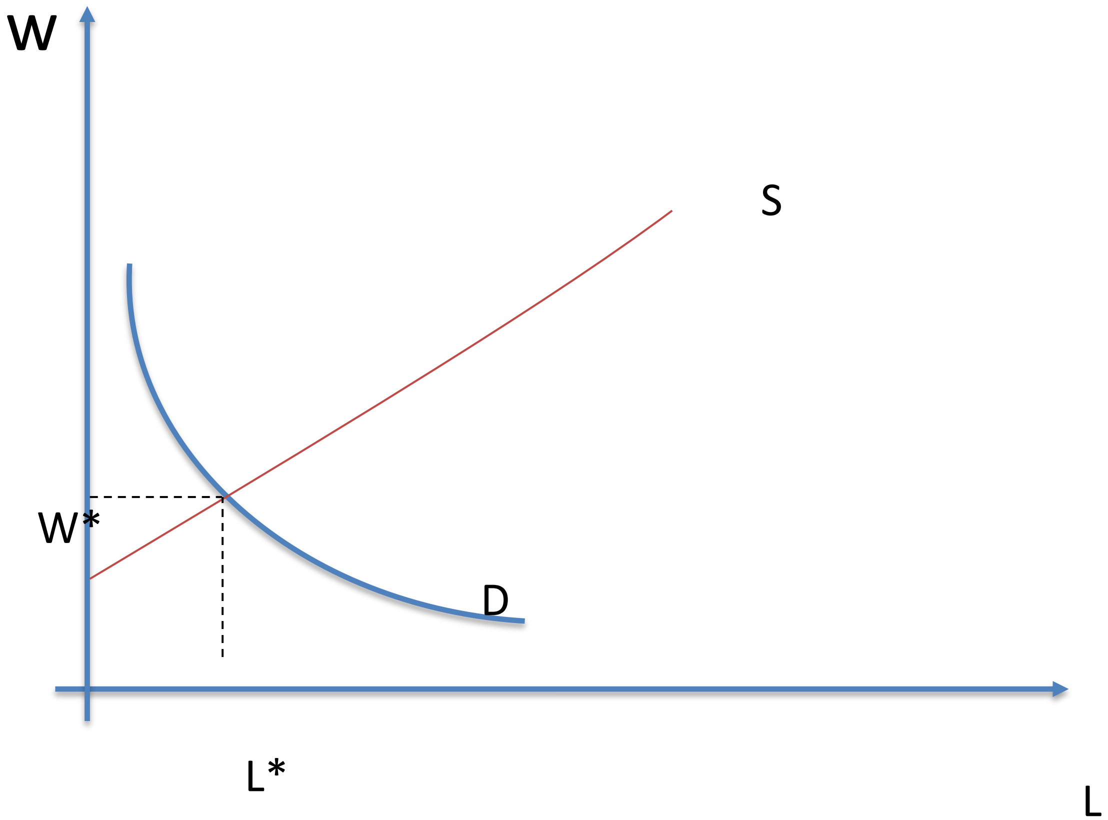
- Individual labor supply:
  - $L(w)=T-N(w)$
- Suppose 200 workers. Then market labor supply:
  - $L^s=200L(w)$

Labor demand curve

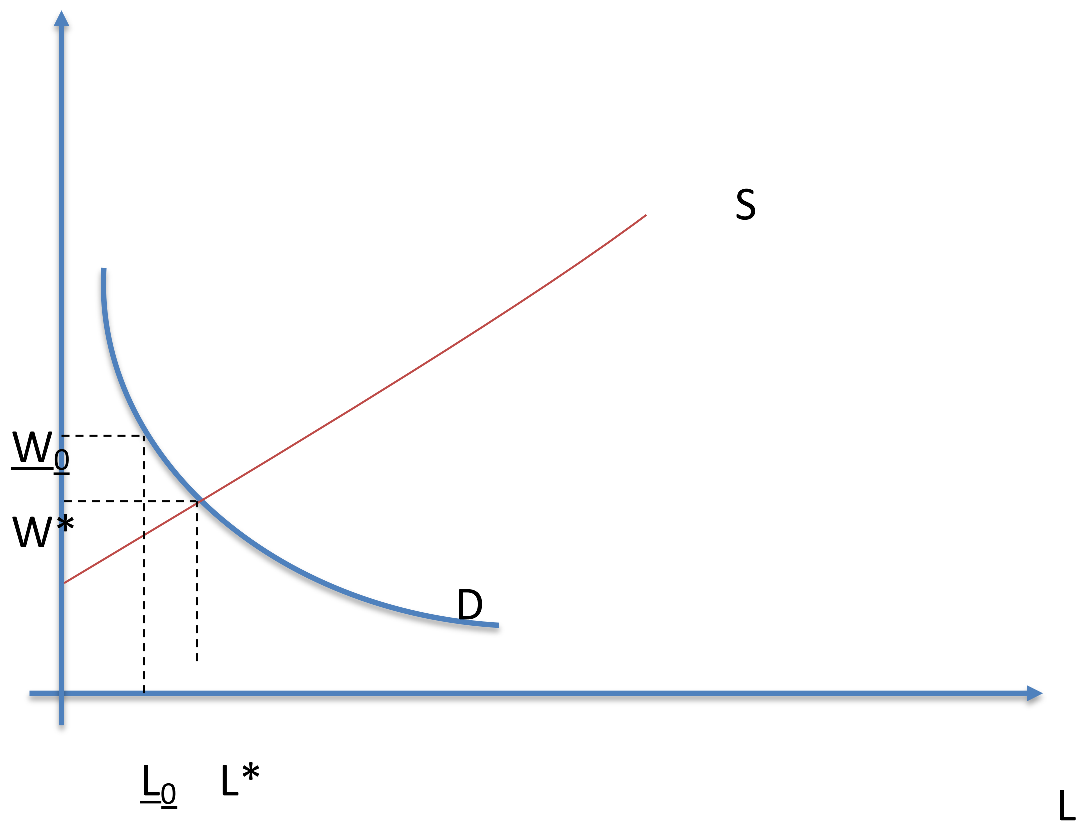
- Individual firm labor demand:
- SET MRL=MCL
  - MCL=w
  - MRL=$P x MP_L$
  - W= $P x MP_L$
  - $P=MC=w/MP_L$
    - L(w)=f(w)
- Suppose 100 firms. Then market labor demand:
  - $L^D=100L(w)$

Labor market equilibrium

- Set $L^s = L^D$ to find the equilibrium



Minimum wage



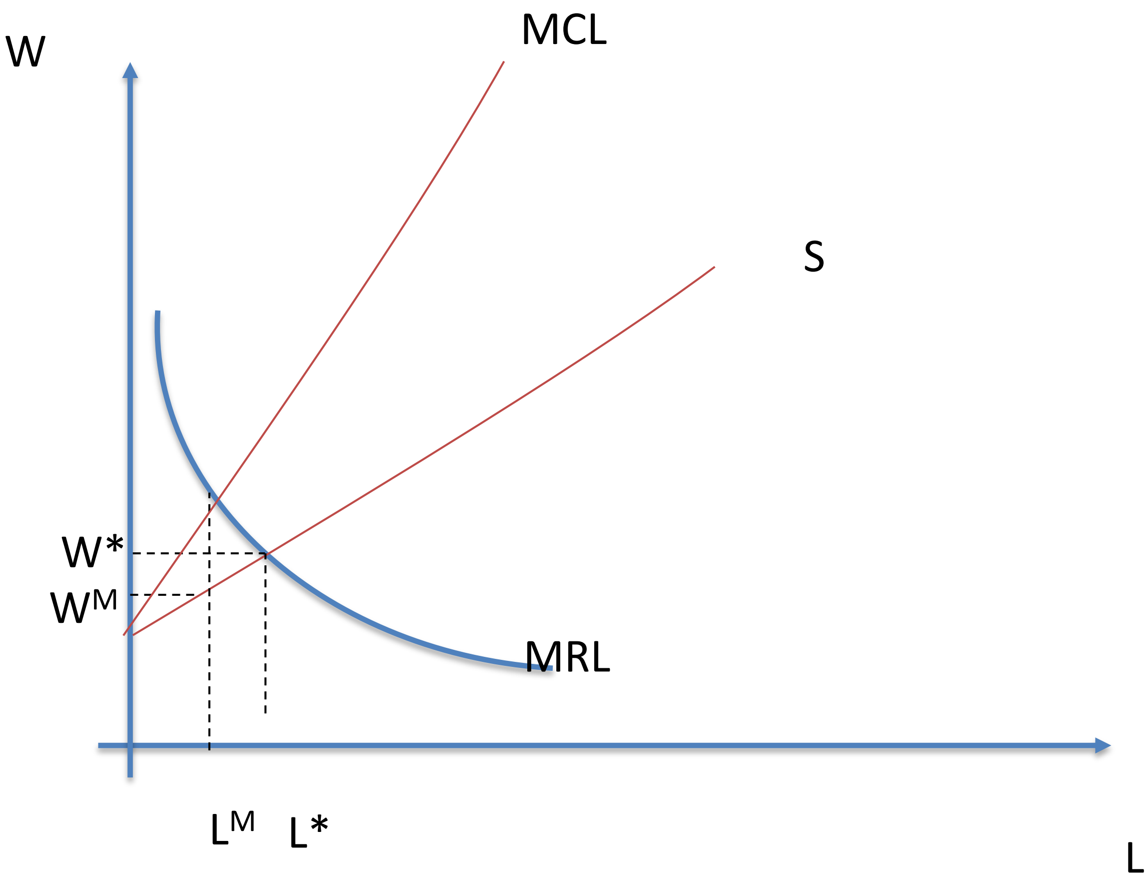
Monopsony in labor market

- Monopsony: many sellers and one buyer
- Often the labor market is a monopsony
  - Market power concentrated among firms
  - Firms faxce an increasing MCL

Monopsony in labor market

- Monopsony: many sellers and one buyer
- Often the labor market is a monopsony
  - Market power concentrated among firms
  - Firms face an increasing MCL
    - Suppose that the (inverse) labor market supply function is W=20+L/100.
      - C=(20+L/100)×L
      - MLC=20+L/50
    - Suppose MLR=80-L/100

Monopsony in labor market



Monopsony in labor market